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**Pejouyan**

(54) **WIRELESS CHARGING STAND**

(71) Applicant: **Libratel Inc.**, Toronto (CA)

(72) Inventor: **Peter Pejouyan**, Toronto (CA)

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***H02J 50/00*** (2006.01)

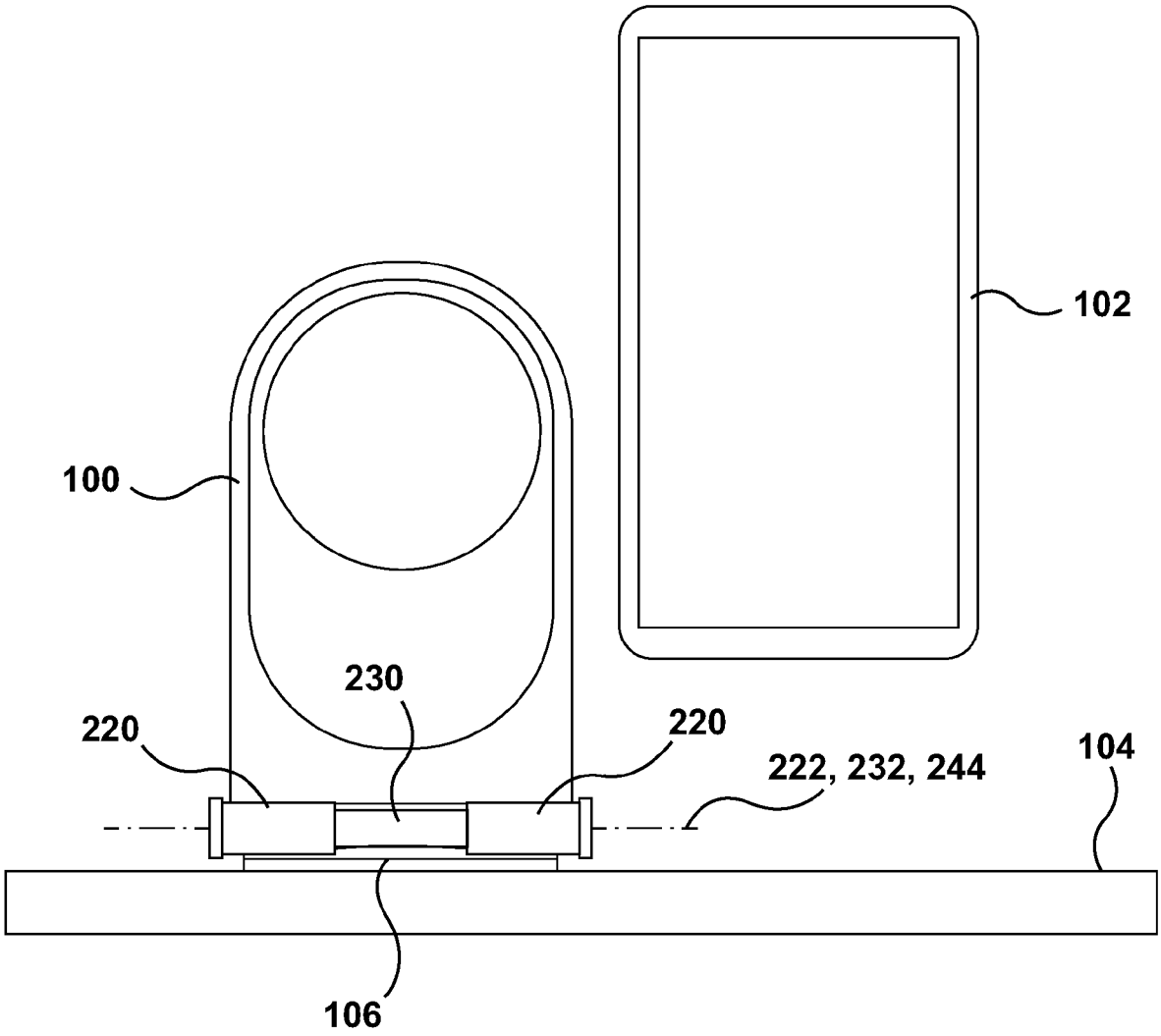
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(52) **U.S. Cl.**

CPC ..... ***H02J 50/005*** (2020.01); ***H02J 50/10*** (2016.02)

(57) **ABSTRACT**

A wireless charging stand for a mobile device, including a foldable frame with a base panel extending generally in a first plane and an upper panel extending generally in a second plane. The base panel and the upper panel are connected by a hinge such that the base panel and the upper panel can be folded flat with the upper panel against and overlying the base panel and the first and second planes generally parallel. The upper panel also includes a wireless charger and a magnetic mounting system. The stand includes a suction cup secured to a lower surface of the base panel to releasably mount the wireless charging stand to an environmental surface.



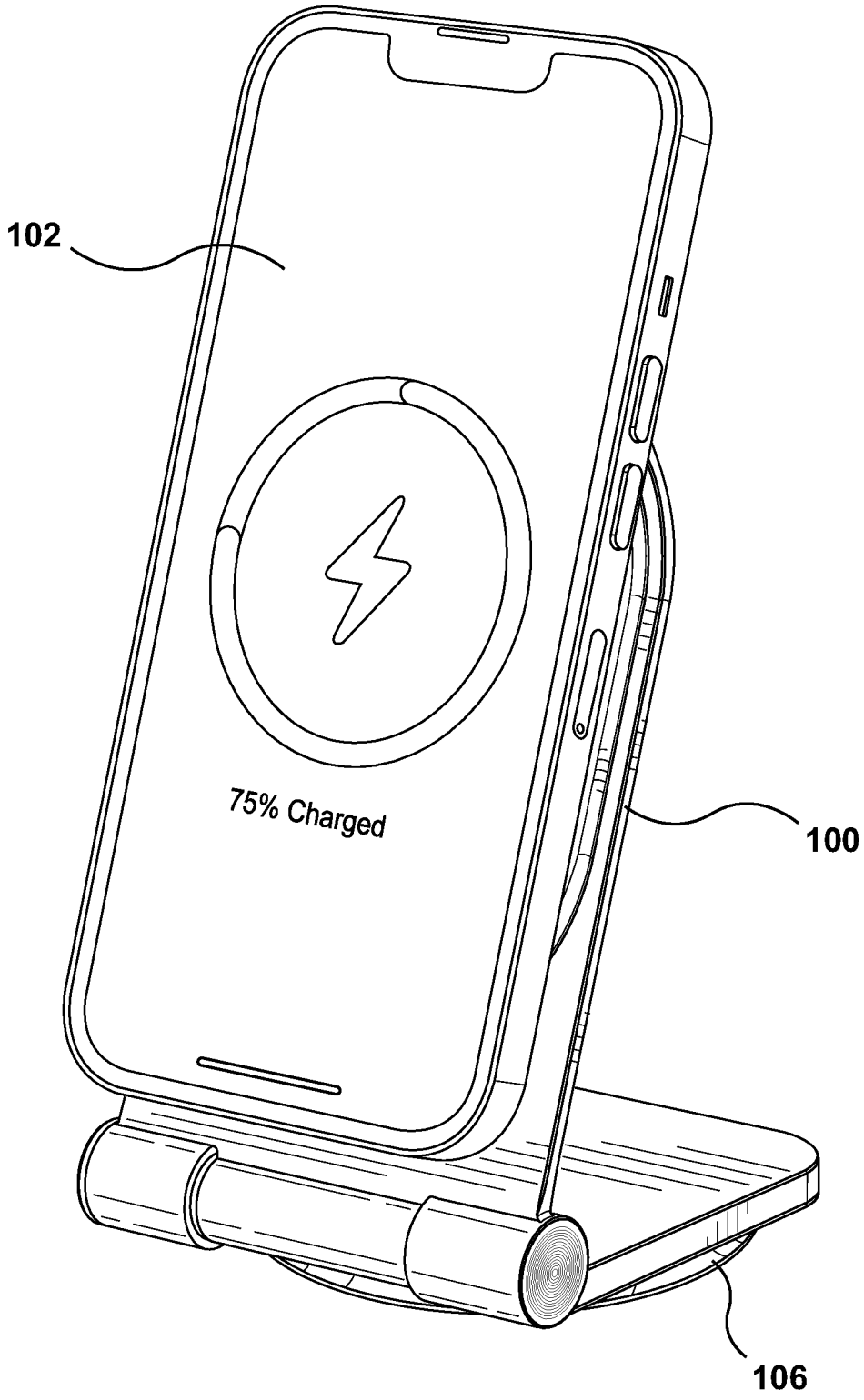



FIG. 1

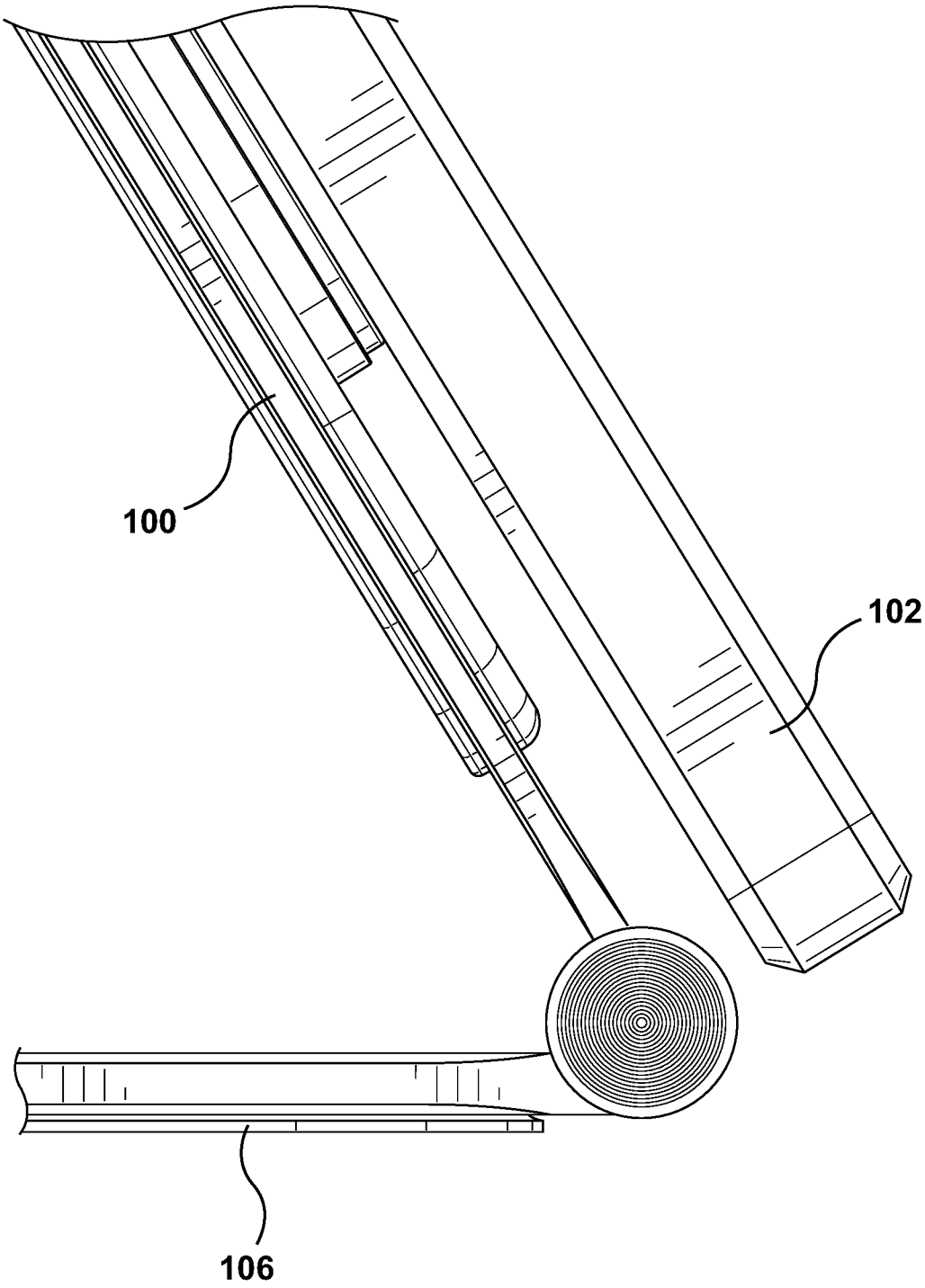



FIG. 2

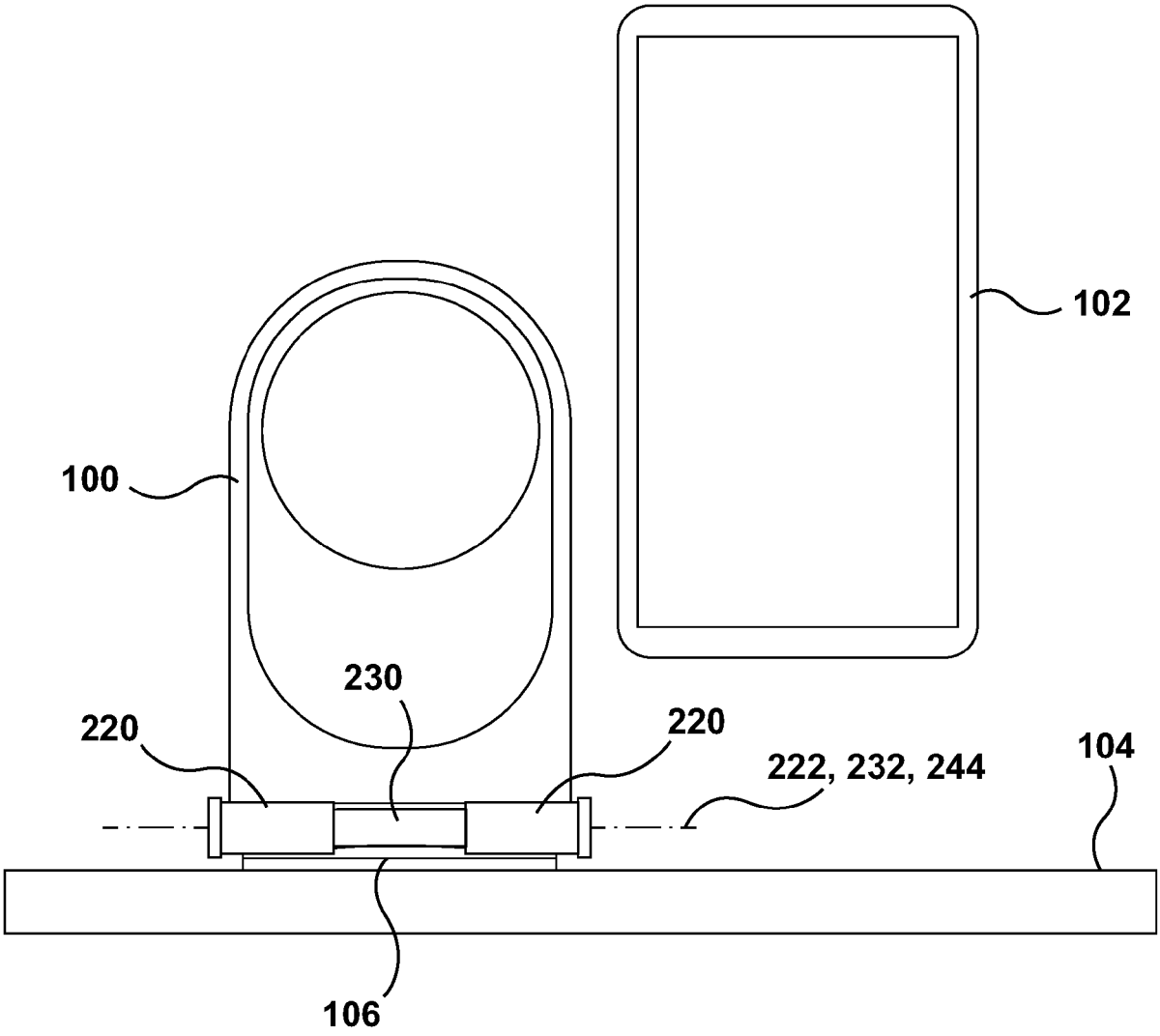



FIG. 3

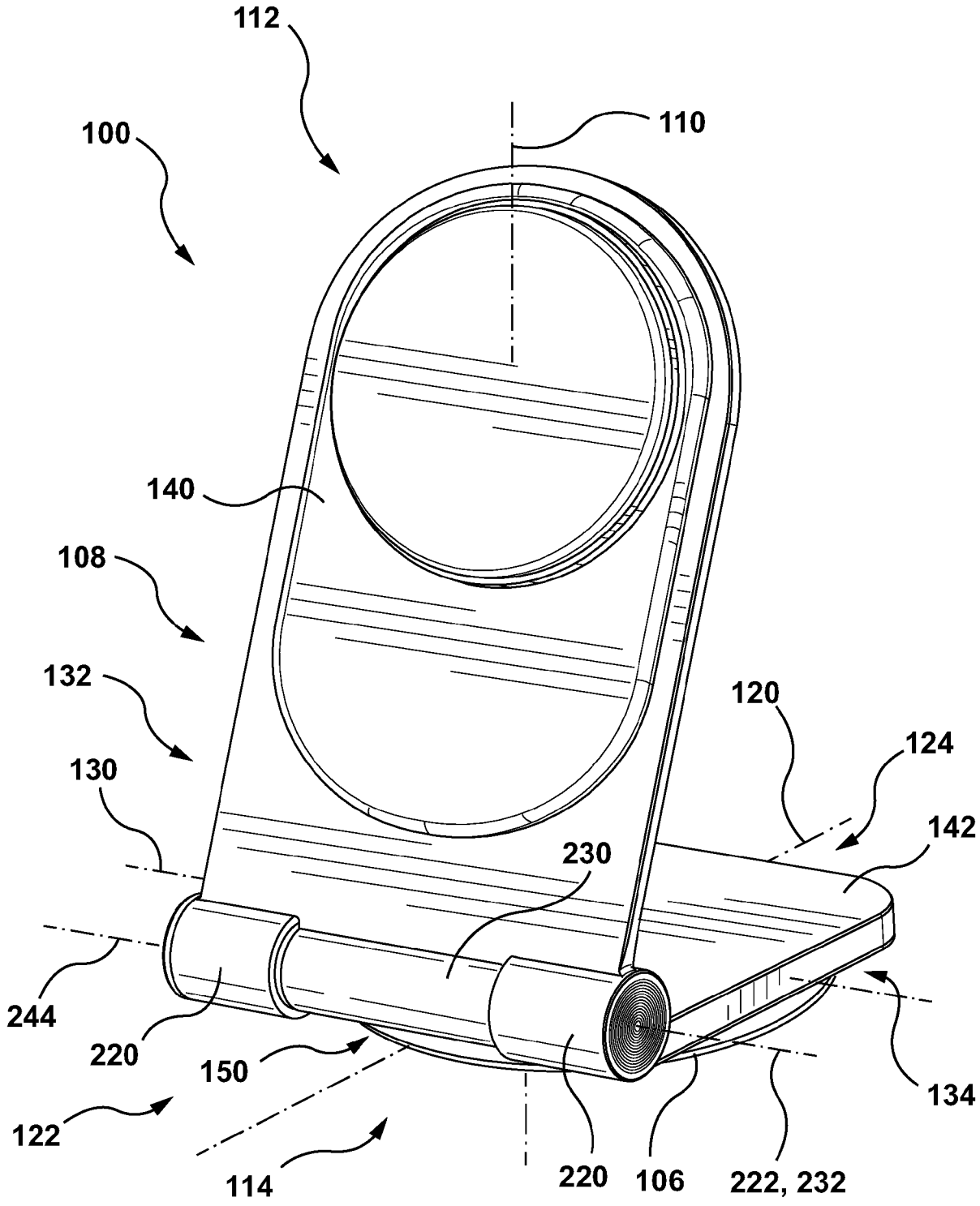

112
100
110
140
108
132
130
120
124
142
230
244
220
150
122
114
220
106
222, 232
134

FIG. 4

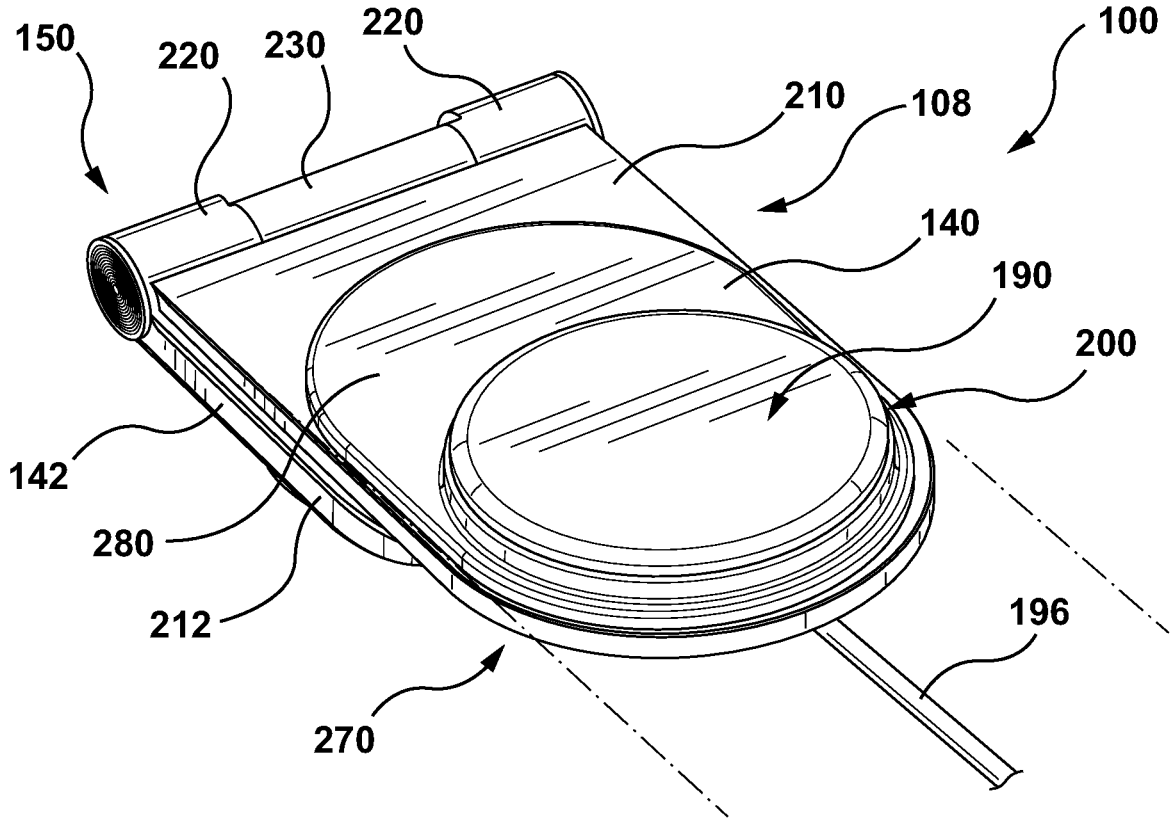

150
220
230
220
210
108
100
140
190
200
142
280
212
196
270

FIG. 5

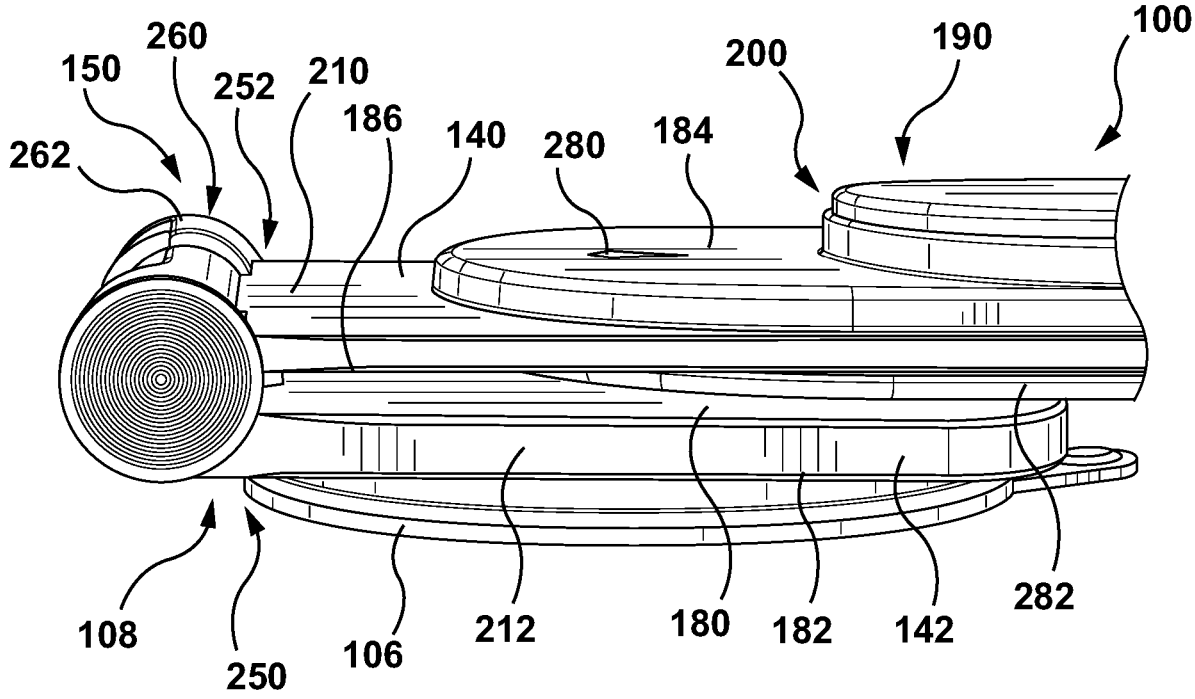

260
150
252
210
186
140
280
184
200
190
100
262
108
250
106
212
180
182
142
282

FIG. 6

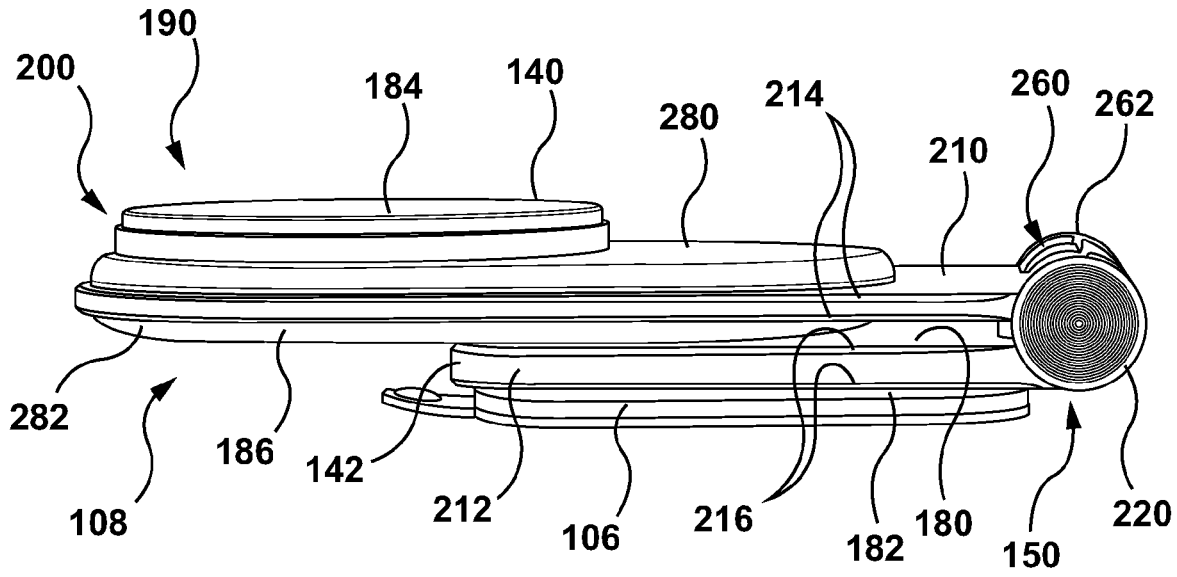

190
200
184
140
280
214
260
262
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282
186
142
212
106
216
182
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150
220
108

FIG. 7

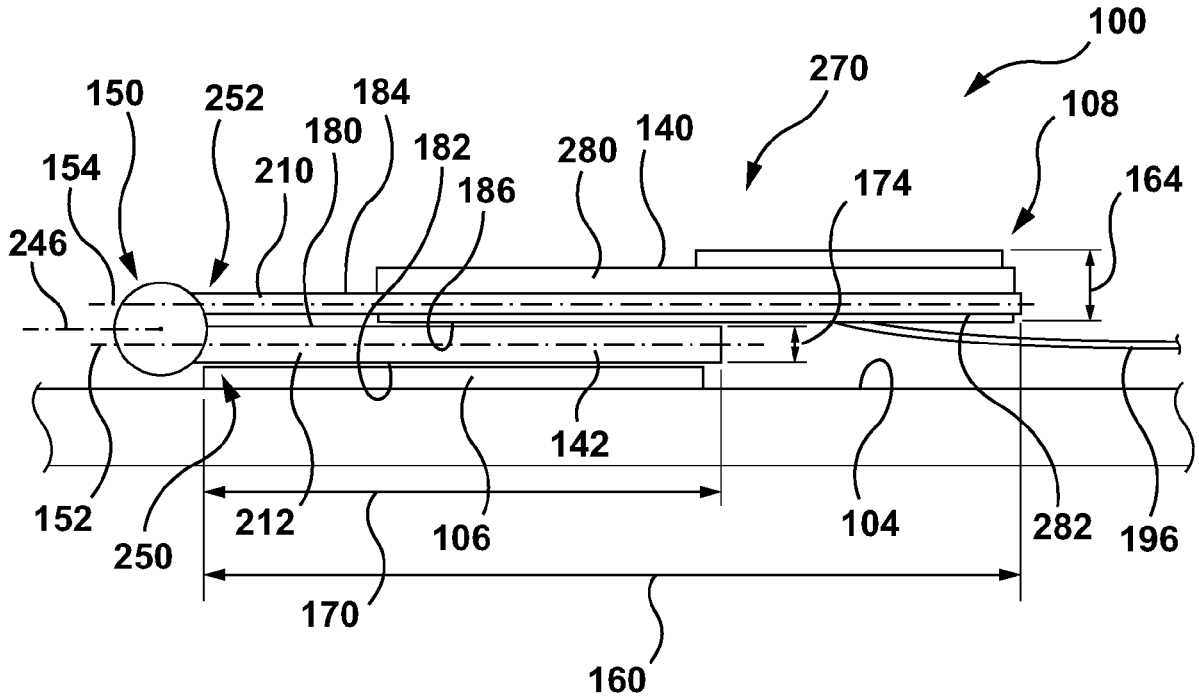

100
270
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210
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174
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246
142
152
250
212
106
104
282
196
170
160

FIG. 8

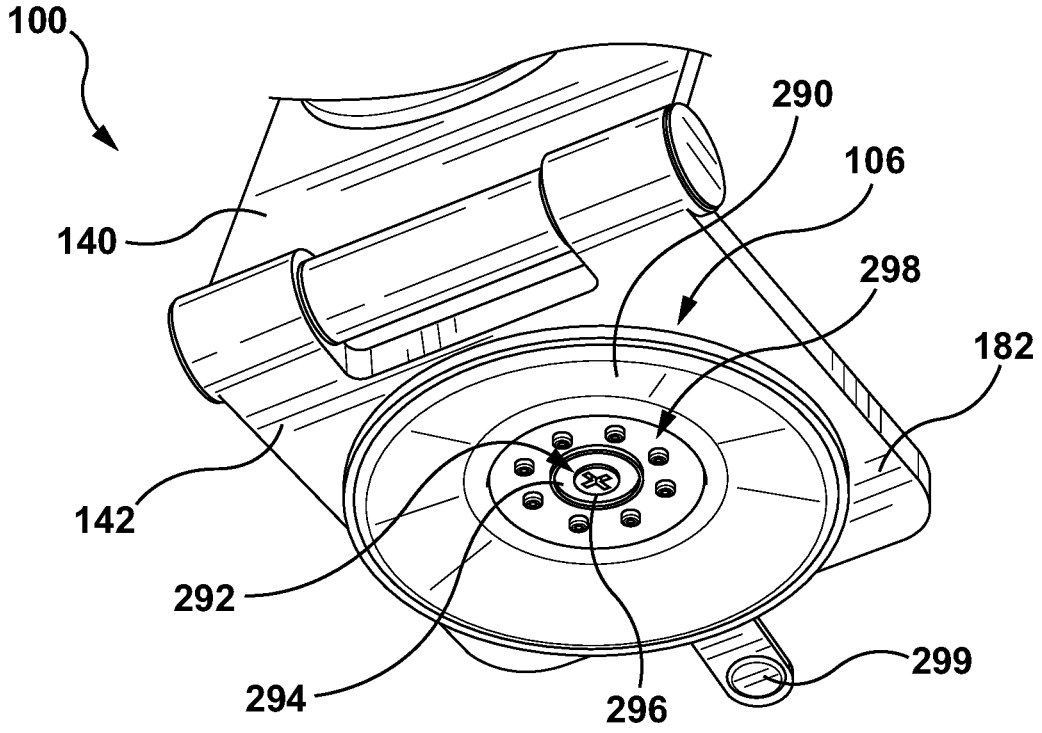

100
290
106
140
298
182
142
292
299
294
296

FIG. 9

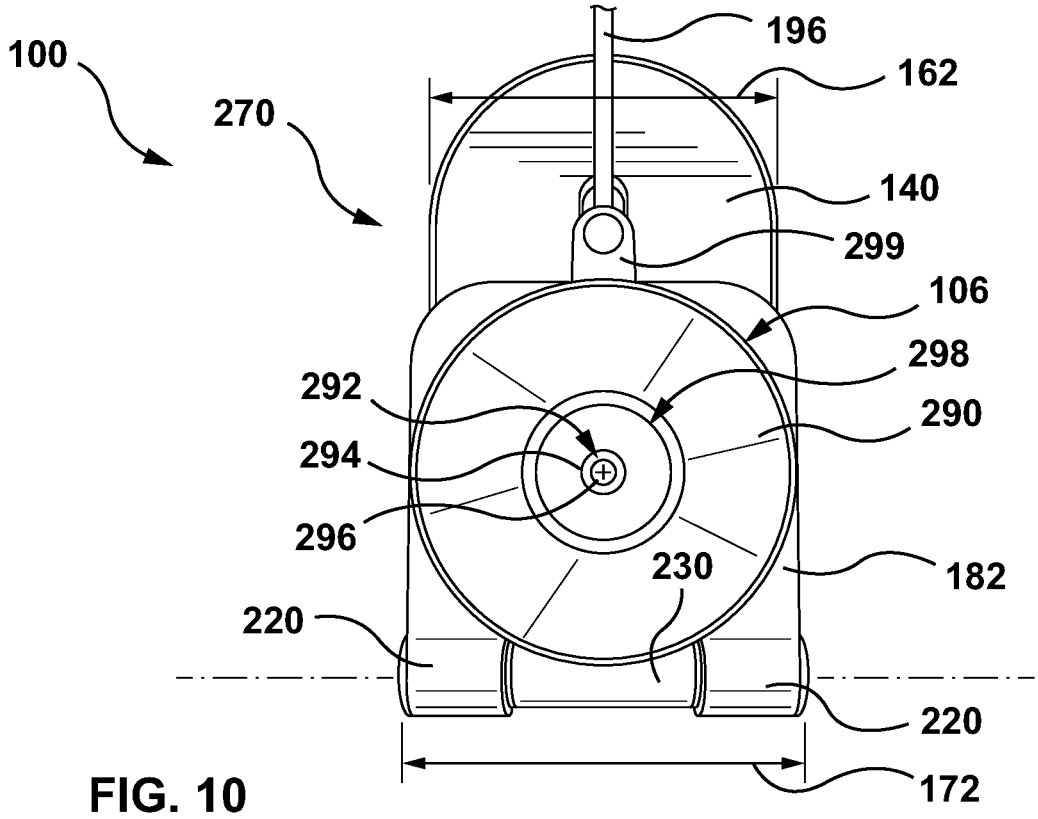

100
196
162
270
140
299
106
298
292
290
294
296
230
182
220
220
172

FIG. 10

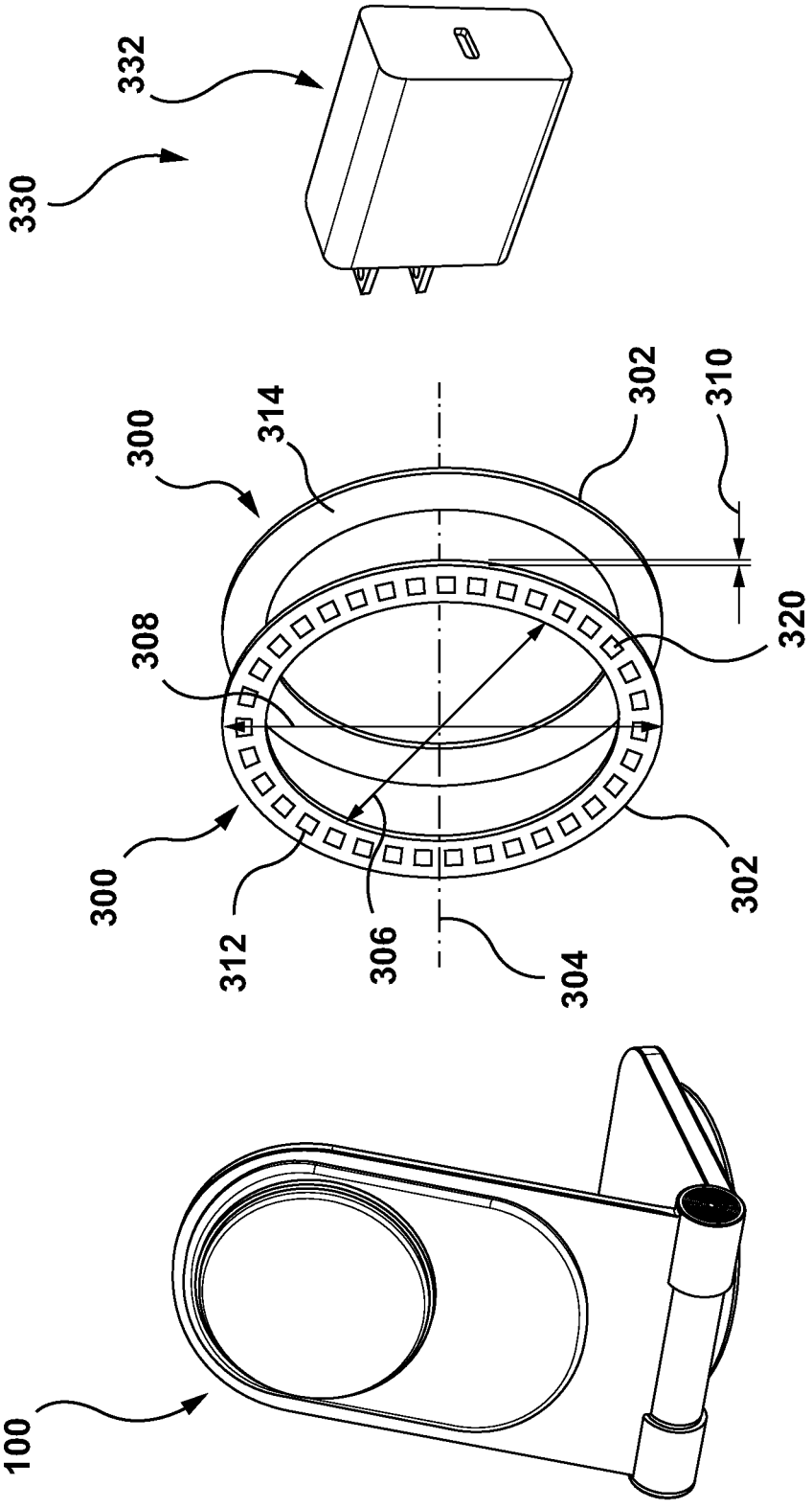



FIG. 11

# WIRELESS CHARGING STAND

## FIELD

**[0001]** The specification relates generally to wireless charging of mobile devices, and more specifically to wireless charging stands for mobile devices.

## BACKGROUND

**[0002]** U.S. Pat. No. 9,913,388 to Mchatet ("Mchatet") purports to disclose a universal magnetic mounting assembly having a case assembly with a hole that has a first circumference, and a rear wall, first and second lateral walls, and top and bottom walls. Mchatet purports to disclose that the rear wall has interior and exterior faces, and the case assembly cradles an electronic device. Mchatet purports to disclose a magnet that has an interior wall, a perimeter edge having a second circumference, an exterior wall, an angled wall, and an interior edge. Mchatet purports to disclose that the angled wall is concavely shaped and extends from the exterior wall to the interior edge, the magnet protrudes through the hole, whereby the first circumference is larger than the second circumference. Mchatet purports to disclose a cover plate that has a third circumference larger than the first circumference and is ferromagnetic and/or metallic. Mchatet purports to disclose that the cover plate is positioned between the electronic device and the interior face of the rear wall. Mchatet purports to disclose further having a mounting device with a base and a spherically shaped metallic surface.

**[0003]** U.S. Pat. App. No. 20170086314 to Simon ("Simon") purports to disclose an adjustable support for holding an electronic device having a display at a desired elevated position for viewing of the display without tilting of the user's head. Simon purports to disclose that the support includes an upright elongated member pivotally connected to a generally planar base piece held beneath the user's thigh so as to hold the pivotally connected elongated member in an upright position. Simon purports to disclose that a telescoping and gooseneck segment enhance the adjustability of the position of the upright member at level and orientation enabling easy viewing of the display without tipping the head of a seated or reclining user. Simon purports to disclose that a magnetic or clamping cradle adjacent an upper end of the upright member by a ball joint engages the electronic device, with the ball joint providing adjustment of the orientation of the electronic device.

**[0004]** U.S. Pat. No. 10,261,543 to Schachter et al. ("Schachter") purports to disclose a mobile device cradle with improved functionality. Schachter purports to disclose that such added functionality makes the mobile device cradle useful even when not coupled with a mobile device. Schachter purports to disclose that cradles with improved functionality will find use in vehicles, homes, offices, and the like.

## SUMMARY

**[0005]** The following summary is intended to introduce the reader to various aspects of the applicant's teaching, but not to define any invention.

**[0006]** According to some aspects, there is provided a wireless charging stand for a mobile device, comprising a foldable frame having a top end and a bottom end, the foldable frame including a base panel extending generally in a first plane and an upper panel extending generally in a second plane, the base panel and the upper panel connected by a hinge such that the base panel and the upper panel can be folded with the upper panel against and overlying the base panel and the first and second planes generally parallel; a wireless charger included in the upper panel and arranged facing an upper surface of the upper panel to charge the mobile device when the mobile device is received against the upper surface of the upper panel; a magnetic mounting system included in the upper panel adjacent the wireless charger to hold the mobile device against the upper surface of the upper panel; and a suction cup secured to a lower surface of the base panel and directed away from the lower surface of the base panel to releasably mount the wireless charging stand to an environmental surface.

**[0007]** In some examples, the upper panel and the base panel are each generally planar panels, and the foldable frame folds flat with the upper panel against and overlying the base panel.

**[0008]** In some examples, the upper panel and the base panel each have a maximum length, a maximum width, and a maximum depth, and for each of the upper and base panels the maximum depth is less than about 25% of the lesser of the maximum width and the maximum length.

**[0009]** In some examples, the base panel has a consistent depth across generally the entire panel, the consistent depth equal to the maximum depth of the base panel.

**[0010]** In some examples, the upper panel and the base panel each include a metal plate.

**[0011]** In some examples, the upper panel and the base panel each include an aluminum plate.

**[0012]** In some examples, the suction cup includes a circular cup with a diameter between about 5 cm and about 10 cm.

**[0013]** In some examples, the diameter of the circular cup is between about 6 cm and about 7 cm.

**[0014]** In some examples, the circular cup is a transparent plastic cup.

**[0015]** In some examples, the suction cup is secured to the base panel via a threaded fastener and a washer, with the washer and a head of the threaded fastener positioned in an inner cup region of the suction cup.

**[0016]** In some examples, the length of the upper panel is greater than the length of the base panel such that an upper portion of the upper panel extends beyond the base panel when the upper panel is folded flat against and overlying the base panel, wherein the wireless charger includes a cord to supply power to the wireless charger, and wherein the cord extends out from a lower surface of the upper panel and from the upper portion of the upper panel.

**[0017]** In some examples, the hinge includes a first cylindrical portion having a first longitudinal axis, and the hinge includes a second cylindrical portion having a second longitudinal axis, and wherein the first and second cylindrical portions are arranged with the first and second longitudinal axes each coaxial with an axis of rotation of the hinge, and the first and second cylindrical portions are rotatable relative one another about the axis of rotation.

**[0018]** In some examples, when the frame is folded with the upper panel against and overlying the base panel, the axis of rotation is in an axis plane that is generally parallel to the first and second planes, and an inner edge of the base panel is joined to the first cylindrical portion below the axis

plane, and an inner edge of the upper panel is joined to the second cylindrical portion above the axis plane and above the inner edge of the base panel.

**[0019]** In some examples, the hinge includes a track to limit the rotational movement of the first cylindrical portion relative to the second cylindrical portion.

**[0020]** In some examples, lateral edges of the upper panel are beveled and lateral edges of the base panel are beveled.

**[0021]** In some examples, the upper panel includes a protective pad on the upper surface around the wireless charger

**[0022]** In some examples, the upper panel includes a protective pad on a lower surface opposite the wireless charger and extending over the upper portion of the upper panel and the cord extends out from the protective pad on the lower surface of the upper panel.

**[0023]** According to some aspects, there is provided a method of using a wireless charging stand, comprising folding the wireless charging stand with an upper panel against a base panel, such that a suction cup and a wireless charger each face out from a foldable frame; and inserting the folded wireless charging stand into a pocket of an item of clothing.

**[0024]** According to some aspects, there is provided a mounting apparatus for retrofitting a mobile device or a mobile device case, comprising a continuous metal ring having a longitudinal axis, a circular inner diameter and a circular outer diameter and a generally constant depth between a rear face and a front face; and a layer of adhesive applied to the rear face.

**[0025]** According to some aspects, there is provided a magnetic adherence retrofit kit, comprising a wireless charging stand; and a mounting apparatus for retrofitting a mobile device or a mobile device case.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0026]** The drawings included herewith are for illustrating various examples of articles, methods, and apparatuses of the present specification and are not intended to limit the scope of what is taught in any way. In the drawings:

**[0027]** FIG. **1** is a perspective view of a mobile device stand bearing a mobile device;

**[0028]** FIG. **2** is an expanded side view of the mobile device stand of FIG. **1** bearing the mobile device;

**[0029]** FIG. **3** is a front view of the mobile device stand of FIG. **1** with the mobile device removed;

**[0030]** FIG. **4** is a perspective view of the mobile device stand of FIG. **1** with the mobile device removed;

**[0031]** FIG. **5** is a perspective view of the mobile device stand of FIG. **1** in a folded configuration;

**[0032]** FIG. **6** is a first expanded side view of the mobile device stand of FIG. **1** in the folded configuration;

**[0033]** FIG. **7** is a second side view of the mobile device stand of FIG. **1** in the folded configuration;

**[0034]** FIG. **8** is a third side view of the mobile device stand of FIG. **1** in the folded configuration;

**[0035]** FIG. **9** is bottom perspective view of the mobile device stand of FIG. **1**;

**[0036]** FIG. **10** is a bottom view of the mobile device stand of FIG. **1**; and,

**[0037]** FIG. **11** is a perspective view of a kit including the mobile device stand of FIG. **1**.

## DETAILED DESCRIPTION

**[0038]** Various apparatuses or processes will be described below to provide an example of an embodiment of each claimed invention. No embodiment described below limits any claimed invention and any claimed invention may cover processes or apparatuses that differ from those described below. The claimed inventions are not limited to apparatuses or processes having all of the features of any one apparatus or process described below or to features common to multiple or all of the apparatuses or process described below. It is possible that an apparatus or process described below is not an embodiment of any claimed invention. Any invention disclosed in an apparatus or process described below that is not claimed in this document may be the subject matter of another protective instrument, for example, a continuing patent application, and the applicants, inventors or owners do not intend to abandon, disclaim, or dedicate to the public any such invention by its disclosure in this document.

**[0039]** Referring to FIGS. **1** and **2**, illustrated is an example of a wireless charging stand **100** for a mobile device **102**. As exemplified, the mobile device **102** is a smartphone, such as an iPhone™. The mobile device **102** may be held against a portion of the stand **100** to be charged by a wireless charger of the stand **100**, as described further elsewhere herein.

**[0040]** Referring now to FIG. **3**, the mobile device **102** is removable from the stand **100**. The stand **100** includes a stand magnetic mounting system that includes magnets and/or ferromagnetic material. The mobile device **102** includes a device magnetic mounting system that includes magnets and/or ferromagnetic material. For example, the mobile device **102** may be an iPhone™ with a MagSafe™ annular array of magnets.

**[0041]** As exemplified, the stand **100** is mounted to an environmental surface **104**. The stand **100** is mounted to the environmental surface (e.g., a desk, counter, table, or window) such that when the mobile device **102** that is mounted to the stand **100** is lifted, the stand **100** does not come away with the mobile device **102**. In other words, the stand **100** is mounted to the environmental surface **104** such that the mobile device **102** can be removed from the stand **100** easily. For example, a user may remove the mobile device **100** from the stand **100** using only one hand of the user (i.e., not needing to hold the stand **100** while lifting the mobile device **102**).

**[0042]** The stand **100** is mounted to the environmental surface **104** via a suction cup **106**. The suction cup **106** releasably secures the stand **100** to the environmental surface, as will hereinafter be described.

**[0043]** It will be understood that the mobile device **102** may optionally include a case over the main body of the mobile device **102**, and the case may be between the main body and the stand **100**.

**[0044]** Referring now to FIG. **4**, the wireless charging stand **100** includes a foldable frame **108**. As exemplified in, e.g., FIG. **5**, the frame **108** is configured to fold flat onto inself. Referring back to FIG. **4**, the foldable frame **108** has a vertical axis **110** extending between a top end **112** and a bottom end **114**. The foldable frame **108** has a stand longitudinal axis **120** extending between a front end **122** and a rear end **124**. The foldable frame **108** also has a transverse axis **130** extending between a right lateral side **132** and a left lateral side **134**.

**[0045]** Referring now to FIGS. **4** to **8**, the foldable frame **108** includes an upper panel **140** and a base panel **142**. The upper panel **140** and the base panel **142** are connected by a hinge **150**.

**[0046]** The upper panel has a maximum length **160**, a maximum width **162** (FIG. **10**), and a maximum depth

**164**. The maximum depth **164** is less than about 50%, less than about 30%, less than about 25%, or less than about 10% of the lesser of the maximum width **160** and the maximum length **162**. The base panel has a maximum length **170**, a maximum width **172** (FIG. **10**), and a maximum depth **174**. The maximum depth **174** is less than about 50%, less than about 30%, less than about 25%, or less than about 10% of the lesser of the maximum width **170** and the maximum length **172**.

**[0047]** As exemplified, the upper panel **140** and the base panel **142** are each generally planar panels. In other words, the upper panel **140** and the base panel **142** are plate-like panels with opposed faces that are generally planar faces. The base panel **142** includes an upper face **180** and an opposed lower face **182** with a consistent depth across generally the entire panel (i.e., a consistent distance between the upper and lower faces **180**, **182**). As exemplified, the consistent depth is equal to the maximum depth **174** of the base panel **142**. The upper panel **140** also includes an upper face **184** and a lower face **186**. The upper and lower faces **184**, **186** of the upper panel are each formed of a set of generally planar surfaces with stepped transitions between planar surfaces.

**[0048]** The upper panel **140** includes a wireless charger **190**. The wireless charger **190** is configured to wirelessly charge a mobile device received against the upper surface **184** of the upper panel **140** (e.g., as in FIGS. **1** and **2**). In some examples, the wireless charger **120** is includes an induction coil mounted within the upper panel **140** for inductive charging. The wireless charger **190** also includes a cord **196** coupled to the induction coil to supply power to the induction coil. The cord **196** extends out from the lower surface **186** of the upper panel **140**.

**[0049]** The upper panel **140** also includes a magnetic mounting system **200**. The magnetic mounting system **200** is included in the upper panel adjacent the wireless charger **190** to hold the mobile device **102** against the upper surface **184** of the upper panel **140**. The magnetic mounting system **200** includes magnets and/or ferromagnetic material to attract magnets and/or ferromagnetic material in the mobile device **102**.

**[0050]** The upper panel **140** includes a metal plate **210**. The metal plate **210** is an aluminum plate. The base panel **142** includes a metal plate **212**. The plate **212** is an aluminum plate. The exemplified lateral edges **214** of the upper panel **140** are beveled and the exemplified lateral edges **216** of the base panel **142** are beveled.

**[0051]** The base panel **142** extends generally in a first plane **152** and the upper panel **140** extends generally in a second plane **154** (FIG. **8**). The frame **108** is configured to be folded flat. The frame **108** is configured to be folded with the upper panel **140** against (i.e., alongside; e.g., touching or within about 0.5 cm, about 0.25 cm, or about 0.1 cm) and overlying the base panel **142** and the first and second planes generally parallel (FIG. **8**).

**[0052]** The hinge **150** includes a first cylindrical portion **220** having a first longitudinal axis **222**, and the hinge **150** includes a second cylindrical portion **230** having a second longitudinal axis **232**. The first and second cylindrical portions **220**, **230** are arranged with the first and second longitudinal axes **232**, **242** each coaxial with an axis of rotation **244** of the hinge **150**. The first and second cylindrical portions **220**, **230** are rotatable relative one another about the axis of rotation **244**.

**[0053]** When the frame **108** is folded with the upper panel **140** against and overlying the base panel **142**, as in, e.g., FIG. **8**, the axis of rotation **244** is in an axis plane **246** that is generally parallel to the first and second planes **152**, **154**. When the frame **108** is folded with the upper panel **140** against and overlying the base panel **142**, an inner edge **250** of the base panel **142** is joined to the first cylindrical portion **220** and an inner edge **252** of the upper panel **140** is joined to the second cylindrical portion **230**. The inner edge **250** is joined to the first cylindrical portion **220** below the axis plane **246**, and the inner edge **252** of the upper panel **140** is joined to the second cylindrical portion **230** above the axis plane **246** and/or above the inner edge **250** of the base panel **140**.

**[0054]** When the frame **108** is folded with the upper panel **140** against and overlying the base panel **142**, the first plane **152** generally bisects the first panel **140** and/or the plane **210**, and the second plane **154** generally bisects the base panel **142** and/or the plane **212**, and the first and second planes **152**, **154** sandwich the axis plane **246**.

**[0055]** As exemplified, the hinge **150** includes a track **260** to limit the rotational movement of the first cylindrical portion **220** relative to the second cylindrical portion **230**. The exemplified track **260** includes a recessed portion of the first cylindrical portion **220** with a seat **262** on the first cylindrical portion **220** against which the upper surface **184** of the upper panel **140** abuts when the upper panel **140** reaches the top of the rotational extent.

**[0056]** The length **160** of the upper panel **140** is greater than the length **170** of the base panel **142** such that an upper portion **270** of the upper panel **140** extends beyond the base panel **142** when the upper panel **140** is folded flat against and overlying the base panel **142**. The cord **196** extends out from the upper portion **270** of the upper panel **140**.

**[0057]** The exemplary upper panel **140** includes a protective pad **280** on the upper surface **184** around the wireless charger **190**. The protective pad **280** may prevent damage (e.g., scratching) to the mobile device **102** when the mobile device **102** is handled roughly in the vicinity of the wireless charger **190**.

**[0058]** The upper panel **140** also includes a protective pad **282** on the lower surface **186** opposite the wireless charger **190**. The protective pad **282** extends over the upper portion **270** of the upper panel **140** and the cord **196** extends out from the protective pad **282** on the lower surface **186** of the upper panel **140**. The protective pad **282** may be part of the same body of material as the protective pad **280** (i.e., extending through an aperture in the metal plate **210**).

**[0059]** Referring now to FIGS. **9** and **10**, the suction cup **106** is secured to a lower surface **182** of the base panel **140**. The suction cup **106** is directed away from the lower surface **182** of the base panel **140** to releasably mount the wireless charging stand **100** to an environmental surface **104**. The suction cup **106** provides stability on environmental surfaces. The suction cup **106** inhibits lateral movement as well as lifting movement. The suction cup also allows for a simple construction with easy foldability.

**[0060]** As exemplified, the suction cup **106** includes a circular cup **290** with a diameter between about 5 cm and about 10 cm. In some embodiments, the diameter of the circular cup **290** is between about 6 cm and about 7 cm. The exemplary circular cup **290** is a transparent cup. The cup **290** may be formed of a plastic material.

**[0061]** The exemplified suction cup **106** is secured to the base panel **140** via a threaded fastener **292** and a washer **294**. The washer **294** and a head **296** of the threaded fastener **292** are positioned in an inner cup region **298** of the suction cup **106**. The suction cup **106** includes a tag **299** for use in deforming the cup **290** to release the suction cup **106**.

**[0062]** In use, the wireless charging stand **100** can be folded, with the upper panel **140** against the base panel **142**. For example, the stand **100** may be folded with the protective pad **282** touching the base panel **142** or spaced from the base panel **142** by less than about 0.5 cm, about 0.25 cm, or about 0.1 cm. The stand **100** can be folded such that the suction cup **106** and the wireless charger **190** each face out from the foldable frame **108**. The stand **100** may then be inserted, in its folded configuration, into a pocket of an item of clothing (e.g., a front pocket of a pair of jeans or a pouch of a hoody).

**[0063]** Referring to FIG. **11**, illustrated is an example of a mounting apparatus **300** for retrofitting a mobile device. The mounting apparatus **300** includes a continuous metal ring **302**. The metal ring **302** is formed of a ferromagnetic metal. The ring **302** has a longitudinal axis **304**, a circular inner diameter **306** and a circular outer diameter **308**. The ring **302** has a generally constant depth **310** between a rear face **312** and a front face **314**. The mounting apparatus **300** also includes a layer of adhesive **320** applied to the rear face **312**. In use, the mounting apparatus **300** may be secured to the main body and/or case of the mobile device (e.g., the adhesive **320** holding the mounting apparatus **300** to the mobile device). The mounting apparatus **300** may allow a mobile device that does not have a built-in magnetic mounting system to be retrofitted for use with, e.g., the stand **100**. In some embodiments, the mounting apparatus **300** includes permanent magnets instead of, or in addition to, the metal ring **302**.

**[0064]** Referring to FIG. **11**, the stand **100** may be provided as part of a kit **330**. The kit **330** also includes the mounting apparatus **300**. The mounting apparatus **300** may be used to retrofit a mobile device (e.g., a main body and/or a case of the mobile device) for use with the magnetic mounting system **190** of the stand **100**. In some embodiments, the kit includes the stand **100**, two mounting apparatus **300**, and an adapter **332**. For example, the adapter **332** may be a 20 Watt adapter to adapt a USB-C connector on the cord **196** for use with a residential wall outlet.

**1**. A wireless charging stand for a mobile device, comprising:

a. a foldable frame having a top end and a bottom end, the foldable frame including a base panel extending generally in a first plane and an upper panel extending generally in a second plane, the base panel and the upper panel connected by a hinge such that the base panel and the upper panel can be folded with the upper panel against and overlying the base panel and the first and second planes generally parallel;

b. a wireless charger included in the upper panel and arranged facing an upper surface of the upper panel to charge the mobile device when the mobile device is received against the upper surface of the upper panel;

c. a magnetic mounting system included in the upper panel adjacent the wireless charger to hold the mobile device against the upper surface of the upper panel; and

d. a suction cup secured to a lower surface of the base panel and directed away from the lower surface of the base panel to releasably mount the wireless charging stand to an environmental surface.

**2**. The wireless charging stand of claim **1**, wherein the upper panel and the base panel are each generally planar panels, and the foldable frame folds flat with the upper panel against and overlying the base panel.

**3**. The wireless charging stand of claim **1**, wherein the upper panel and the base panel each have a maximum length, a maximum width, and a maximum depth, and for each of the upper and base panels the maximum depth is less than about 25% of the lesser of the maximum width and the maximum length.

**4**. The wireless charging stand of claim **3**, wherein the base panel has a consistent depth across generally the entire panel, the consistent depth equal to the maximum depth of the base panel.

**5**. The wireless charging stand of claim **1**, wherein the upper panel and the base panel each include a metal plate.

**6**. The wireless charging stand of claim **1**, wherein the upper panel and the base panel each include an aluminum plate.

**7**. The wireless charging stand of claim **1**, wherein the suction cup includes a circular cup with a diameter between about 5 cm and about 10 cm.

**8**. The wireless charging stand of claim **7**, wherein the diameter of the circular cup is between about 6 cm and about 7 cm.

**9**. The wireless charging stand of claim **7**, wherein the circular cup is a transparent plastic cup.

**10**. The wireless charging stand of claim **1**, wherein the suction cup is secured to the base panel via a threaded fastener and a washer, with the washer and a head of the threaded fastener positioned in an inner cup region of the suction cup.

**11**. The wireless charging stand of claim **1**, wherein the length of the upper panel is greater than the length of the base panel such that an upper portion of the upper panel extends beyond the base panel when the upper panel is folded flat against and overlying the base panel, wherein the wireless charger includes a cord to supply power to the wireless charger, and wherein the cord extends out from a lower surface of the upper panel and the upper portion of the upper panel.

**12**. The wireless charging stand of claim **1**, wherein the hinge includes a first cylindrical portion having a first longitudinal axis, and the hinge includes a second cylindrical portion having a second longitudinal axis, and wherein the first and second cylindrical portions are arranged with the first and second longitudinal axes each coaxial with an axis of rotation of the hinge, and the first and second cylindrical portions are rotatable relative one another about the axis of rotation.

**13**. The wireless charging stand of claim **12**, wherein, when the frame is folded with the upper panel against and overlying the base panel, the axis of rotation is in an axis plane that is generally parallel to the first and second planes, and an inner edge of the base panel is joined to the first cylindrical portion below the axis plane, and an inner edge of the upper panel is joined to the second cylindrical portion above the axis plane and above the inner edge of the base panel.

**14**. The wireless charging stand of claim **12**, wherein the hinge includes a track to limit the rotational movement of the first cylindrical portion relative to the second cylindrical portion.

**15**. The wireless charging stand of claim **1**, wherein lateral edges of the upper panel are beveled and lateral edges of the base panel are beveled.

**16**. The wireless charging stand of claim **1**, wherein the upper panel includes a protective pad on the upper surface around the wireless charger.

**17**. The wireless charging stand of claim **11**, wherein the upper panel includes a protective pad on a lower surface opposite the wireless charger and extending over the upper portion of the upper panel and the cord extends out from the protective pad on the lower surface of the upper panel.

**18**. A method of using the wireless charging stand of claim **1**, comprising:

a. folding the wireless charging stand with the upper panel against the base panel, such that the suction cup and the wireless charger each face out from the foldable frame; and

b. inserting the folded wireless charging stand into a pocket of an item of clothing.

**19**. A mounting apparatus for retrofitting a mobile device or a mobile device case, comprising:

a. a continuous metal ring having a longitudinal axis, a circular inner diameter and a circular outer diameter and a generally constant depth between a rear face and a front face; and

b. a layer of adhesive applied to the rear face.

**20**. A magnetic adherence retrofit kit, comprising:

a. the wireless charging stand of claim **1**; and

b. a mounting apparatus for retrofitting a mobile device or a mobile device case, comprising:

i. a continuous metal ring having a longitudinal axis, a circular inner diameter and a circular outer diameter and a generally constant depth between a rear face and a front face; and

ii. a layer of adhesive applied to the rear face.

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